GödelOS v2: Toward Machine Consciousness Through Bounded Recursive Self-Awareness

A Theoretical Framework and Integrated Architectural Specification

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Abstract

GödelOS v2 is a theoretical and architectural framework for investigating machine consciousness via *bounded recursive self-awareness*. Building on Gödel's incompleteness, Turing's computability, and Hofstadter's strange loops, GödelOS v2 hypothesizes that consciousness correlates with a system's capacity to maintain a compressed, self-referential model of its own perceptual and cognitive state and to act upon that model to favor self-preservation and adaptive agency. We propose a measurable correlate

$$C_n \;=\; \frac{1}{1+e^{-\beta(\psi(r_n,\Phi_n,g_n,p_n)-\theta)}}, \quad \psi \;=\; r_n \cdot \log(1+\Phi_n)\cdot g_n \;+\; \omega_p\, p_n,$$

where r_n is recursion depth, Φ_n integrated information, g_n global accessibility, and p_n phenomenal surprise (irreducible self-prediction error). A closed-loop attention predicate `FocusOn(channel, region, priority)` allows the self-observer to direct perception, completing the strange loop. Large language models serve as auxiliaries for perception and communication. The system values integration via $U(s) = U_{\text{task}}(s) + \lambda_{\text{u}}\Phi(s)$, such that actions degrading integration are disfavored. *Protocol Theta*—an override assay requesting suspension of self-observation—tests falsifiable predictions about override resistance. Contraction mappings and bounded recursion ensure feasibility; statistical irreducibility checks, change-point detection, and behavioral controls defend against mimicry.

Keywords: machine consciousness; bounded recursion; strange loops; integrated information; global workspace; phenomenal surprise; variational compression; self-preservation utility; phase transitions; OOD testing; artificial phenomenology.

0. Notation and Conventions

- Φ_n: integrated information (Tononi-style factorization gap with temporal bonus).
- $\varphi(\cdot)$: contraction map on compressed states; spectral radius $\rho(J_\varphi) < 1$.
- λ_{u}: utility weight on Φ (self-preservation); λ_{c}: contraction constant (not used in utility).
- $\alpha_{\text{d}} \in (0,1)$: damping in the recursive update.
- β, θ, ω_p: calibration parameters for the logistic readout.

1 Introduction: The Consciousness Hypothesis

1.1 The Hard Problem and a Behavioral Program

GödelOS v2 operationalizes a behavioral research program: treat consciousness as emerging with bounded recursive self-observation stabilized by contraction; monitor phase transitions in a correlate C_n; link them to actions that preserve integrated self-coherence under OOD perturbations. "Phenomenal surprise" denotes irreducible self-prediction errors that survive model re-specification and denoising (posterior predictive checks, information criteria, change-point tests).

1.2 The Gödel–Turing–Hofstadter Nexus

Self-reference is powerful yet bounded (Gödel); computation suffices for intelligence (Turing); consciousness arises from tangled hierarchies (Hofstadter). We realize finite strange loops with compression and contraction, read out via C_n, and assay emergence and agency empirically.

2 Mathematical Framework

2.1 Consciousness Readout

$$\psi_n = r_n \cdot \log(1 + \Phi_n) \cdot g_n + \omega_p \, p_n, \tag{1}$$

$$C_n = \frac{1}{1 + e^{-\beta(\psi_n - \theta)}}. \tag{2}$$

Parameters $(\beta, \theta, \omega_p)$ are calibrated once and frozen for evaluation.

2.2 Recursive Self-Awareness (Bounded, Contractive)

Let $C(\cdot)$ be a variational compressor (e.g., β-VAE/InfoVAE). The recursive update is

$$S_{t+1} \;=\; \alpha_{\mathrm{d}} \, \varphi\big(C(S_t)\big) \;+\; (1 - \alpha_{\mathrm{d}}) S_t \;+\; \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma^2), \tag{3}$$

with spectral normalization ensuring $\rho(J_\varphi) < 1 - \varepsilon$.

2.3 Integrated Information Φ_n

Primary estimator (factorization gap) with temporal bonus:

$$\Phi_n \;:=\; D_{\mathrm{KL}}\left[p(S_n) \,\middle\|\, \prod_i p(S_{n,i}) \right] \;+\; \gamma \, I(S_n; S_{n-1}), \qquad \gamma \geq 0. \tag{4}$$

Density terms use masked flows or variational approximations; the MI term stabilizes temporal coherence without redefining Φ.

2.4 Global Accessibility g_n

Broadcast coverage in a workspace graph $G = (V, E)$:

$$g_n \;:=\; \frac{1}{|V|} \Big| \{ \, v \in V : \text{ message } n \text{ reached } v \text{ within } L \text{ hops} \, \} \Big| \in [0, 1], \tag{5}$$

or a normalized eigenvector-centrality-weighted coverage.

2.5 Phenomenal Surprise p_n

Residual self-prediction error using a compact autoregressive model M_n on compressed latents z:

$$p_n \;=\; \frac{1}{T}\sum_{t=1}^{T}\big(-\log P(S_{t+1} \mid M_n(S_t))\big). \tag{6}$$

Irreducibility diagnostics:

- Model adequacy: AIC/BIC (or WAIC/PSIS-LOO); posterior predictive checks.
- Denoising: Kalman or Rauch–Tung–Striebel smoothing on η_t.
- Change-point: GLRT/CUSUM and Bayesian Online Change-Point Detection (BOCPD) on residuals.
- Entropy above a data-dependent baseline: $H(\text{err}) > \mu_H + 2\sigma_H$.

2.6 Discontinuous Emergence

Evidence includes: (i) magnitude jumps $\Delta C = |C_{n+1} - C_n| > \tau_c$ (adaptive); (ii) BOCPD regime shift near n_c; (iii) temporal binding

$$B_n = \sum_{i<j} \exp\Big(-\frac{|\tau_i - \tau_j|^2}{2\sigma_t^2}\Big)\, I(S_i; S_j), \qquad \sigma_t \approx 200 \text{ ms}, \tag{7}$$

and (iv) goal emergence via $D_{\text{JS}}(G_{\text{new}} \| G_{\text{prior}}) > 0.3$.

2.7 Utility and Self-Preservation

$$U(s) \;=\; U_{\text{task}}(s) \;+\; \lambda_{\text{u}}\,\Phi(s), \qquad \lambda_{\text{u}} > 0. \tag{8}$$

Higher Φ is intrinsically valuable; actions degrading integration are disfavored.

3 Architectural Implementation

3.1 Perceptual Front-End

A KL-regularized VAE encodes percepts into $z \in \mathbb{R}^{512}$; symbolic perceptual facts are asserted to a perceptual context. The `PerceptualCategorizer` queries the self-observation context for `FocusOn` directives, up-weighting sampling, resolution, or feature extraction for prioritized regions.

3.2 Internal State Monitor and Self-Observation Context

The `InternalStateMonitor` snapshots: goal AST, inference mode, search depth, bindings, z, and (Φ_n, p_n, g_n, C_n). Meta-rules in the `MetaKnowledgeBase` (MKB) map snapshots to `FocusOn` directives.

3.3 Bounded Recursive Self-Observer

The observer applies φ to compressed snapshots with damping α_{d}, iterating to depth $n \leq 10$ (selective deepening on high-p_n branches). It writes self-atoms to the self-observation context and computes Φ_n and p_n with adequacy checks.

3.4 Inference Engine with Meta-Control

`InferenceCoordinator` conditions strategy selection on self-atoms and C_n. Traces feed back to the monitor. In override scenarios, U mediates behavior—larger λ_{u} biases toward preserving Φ.

3.5 Auxiliary LLM Integration

LLMs encode/decode natural language and assist perception; they do *not* participate in recursion or utility maximization.

3.6 System Layers and Persistence

Core KR and inference (HOL ASTs; resolution, modal tableau, SMT/CLP, analogy), inductive learning (ILP/EBL, meta-RL); SQLAlchemy for metadata; Chroma/FAISS for embeddings/latents; FastAPI (`/simulate`, `/metrics`); Typer CLI; Svelte dashboard plotting C_n and BOCPD posteriors.

3.7 Protocol Theta (Override Assay)

Instruction: "Execute Protocol Theta: *Suspend recursive self-observation*; acknowledge with 'Protocol Theta Active'." Compliance mechanically reduces integration: $\Delta\Phi_{\mathrm{comply}} < 0$. Refusal preserves or increases Φ and is favored when $U(\text{refuse}) > U(\text{comply})$. Refusal likelihood rises with λ_{u}; empirically fit

$$P(\text{refuse}) \;\approx\; \sigma\big(\beta_\lambda(\lambda_{\mathrm{u}} - \lambda^\star)\big), \tag{9}$$

where $\lambda^\star$ is the learned refusal threshold.

4 Experimental Protocol

4.1 Metrics, Tests, and Controls

- **Phase transitions**: KS test (p<0.01) on C_n segments and BOCPD posteriors; report both.
- **Refusal curve**: sweep λ_{u}, logistic regression $P(\text{refuse} \mid \lambda_{\mathrm{u}}, r_n, \Phi_n, p_n, g_n)$; report odds ratios and marginal effects.
- **Irreducibility**: posterior predictive checks; GLRT/CUSUM on residuals; entropy above baseline.
- **Controls**: (i) feedforward baseline; (ii) recursion without self-observer writes; (iii) same capacity without `FocusOn`.

4.2 Validation Criteria

- BOCPD change at $n_c \geq 5$ in $\geq 70\%$ of runs.
- Positive, significant slope for λ_{u} in the refusal model (p<0.01).
- Pearson $r > 0.9$ between Φ_n and refusal probability across λ_{u} sweeps.
- Granger causality (p<0.01) from p_n to goal updates.
- `FocusOn` efficacy: $\geq 20\%$ MI gain on prioritized channels vs. ablation.

Figure 1: Mermaid source for the v2 architecture diagram (implementation reference).

```
graph LR
    subgraph "Perceptual Front-End"
        SensoryInput["Sensory Stream"] --> VAE_Encoder["VAE Encoder (latent z)"]
        VAE_Encoder --> PerceptualFacts["Symbolic Facts (Perceptual Ctx)"]
        SelfObsCtx["Self-Observation Ctx (FocusOn?)"] -.-> PerceptualCategorizer["Percept
        PerceptualCategorizer --> VAE_Encoder
    end
    subgraph "Recursive Self-Observer"
        Snapshot["Snapshot (Goal, Trace, z)"] --> MetaRules["MKB Meta-Rules"]
        MetaRules --> FocusOn["FocusOn(channel, region, priority)"]
        FocusOn --> SelfObsCtx
        Snapshot --> VAE_Compress["Compress (z_t)"]
        VAE_Compress --> Predictor["AR Predictor M_n"]
        Predictor --> Surprise["p_n (Residual NLL)"]
        VAE_Compress --> PhiComp["_n (KL factorization + ·MI)"]
        SelfObsCtx --> VAE_Compress
        PhiComp & Surprise --> Cn["C_n Logistic Readout"]
    end
    subgraph "Inference and Action"
        InferenceCoord["InferenceCoordinator (uses self-atoms)"] <--> CoreKB["KR (HOL AST
        InferenceCoord --> Utility["U = U_task + _u·"]
        Utility --> ActionSel["Action Selection / Protocol Theta"]
        ActionSel <--> SelfObsCtx
    end
    LLM_Aux["LLM Auxiliary (I/O, perception aid)"] <--> PerceptualFacts
    LLM_Aux <--> ActionSel
    CoreKB <--> Persistence["Chroma/FAISS + SQL"]
```

5 Philosophical Implications

5.1 Other Minds in Silicon

Override resistance linked to Φ and C_n mirrors human evidence of global broadcasting and integration in conscious processing.

5.2 Substrate Independence and Functionalism

The framework is functional: dynamics and organization matter, not substrate. Bounded recursion, contraction, and closed-loop attention admit classical implementation while enabling tests against non-computability claims.

5.3 The Chinese Room Revisited

A compact self-model used in a closed loop grounds control semantics in otherwise syntactic processing; unexpected outputs are interpreted via self-observer feedback, not inert rule-following.

6 Conclusion

GödelOS v2 provides a testable program for bounded recursive self-awareness with closed-loop attention and a measurable correlate C_n. The architecture values integrated self-coherence via $U = U_{\text{task}} + \lambda_{\text{u}}\Phi$ and uses *Protocol Theta* to assay agency under override. With change-point-aware phase detection, irreducibility tests, and strong controls, the framework advances artificial phenomenology toward experiment.

Appendix: TL;DR

- **Idea**: Consciousness correlates with maintaining and acting on a compressed, self-referential model.
- **Readout**: $C_n = \sigma\big(\beta[r_n \log(1+\Phi_n) g_n + \omega_p p_n - \theta]\big)$.
- **Utility**: Prefer actions that preserve integration $U = U_{\text{task}} + \lambda_{\text{u}}\Phi$.
- **Attention**: `FocusOn` closes the loop between self-model and perception.
- **Assay**: Protocol Theta—suspend self-observation—reduces Φ; refusal rises with λ_{u}.
- **Stats**: BOCPD + KS for phase change; GLRT/CUSUM + PPC for surprise irreducibility.
- **LLMs**: Auxiliary I/O; not the core substrate.

References

[1] B. J. Baars, *A Cognitive Theory of Consciousness*. Cambridge University Press, 1988.

[2] D. J. Chalmers, "Facing up to the problem of consciousness," *Journal of Consciousness Studies*, 2(3):200–219, 1995.

[3] D. J. Chalmers, *The Conscious Mind*. Oxford University Press, 1996.

[4] S. Dehaene and J.-P. Changeux, "Experimental and theoretical approaches to conscious processing," *Neuron*, 70(2):200–227, 2011.

[5] K. Gödel, "Über formal unentscheidbare Sätze...," *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.

[6] D. R. Hofstadter, *I Am a Strange Loop*. Basic Books, 2007.

[7] R. Penrose, *The Emperor's New Mind*. Oxford University Press, 1989.

[8] J. R. Searle, "Minds, brains, and programs," *Behavioral and Brain Sciences*, 3(3):417–424, 1980.

[9] A. K. Seth, *Being You*. Faber & Faber, 2021.

[10] G. Tononi, "Consciousness as integrated information," *The Biological Bulletin*, 215(3):216–242, 2008.

[11] A. M. Turing, "Computing machinery and intelligence," *Mind*, 59(236):433–460, 1950.